

Individual Listener Preferences for Music Mixes Across Listening Contexts: Statistical Analysis and Personalised LLM Prediction

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Abstract— *Music-mix preference is inherently listener-dependent and may also vary with listening context, yet conventional mixing workflows typically optimise a single fixed version for all listeners and situations. This dissertation investigates how preferences for alternative music mixes vary across listeners and functional listening contexts, and whether individual preferences can be predicted computationally. A controlled listening study collected 990 mix-preference ratings across 198 trials from 33 participants, using four songs, five professionally produced alternative mixes per song, and three functional listening contexts. Bayesian multilevel modelling was used to quantify context-, listener-, stimulus-, and acoustic-related variation in preference. Four large language models—GPT-5.5, Claude Sonnet 5, Llama 3.1 70B Instruct, and Centaur—were subsequently evaluated for held-out preferred-mix prediction, both with and without prior-trial ratings and comments from the same listener. The statistical analysis identified systematic contextual and acoustic associations, including lower ratings in the Focus-Motivation context relative to Enjoyment and a negative association between crest factor and preference, while participant- and stimulus-level variance indicated additional preference heterogeneity. Held-out prediction was modest but exceeded the nominal 20% five-choice chance reference for the strongest configurations. Claude Sonnet 5 with participant history achieved 35.9% Top-1 accuracy, while GPT-5.5 with history and the matched mixed-effects predictor each achieved 34.3%. Prior listener evidence increased Top-1 accuracy by 15.2 percentage points for Claude and 11.6 points for GPT-5.5, but produced substantially smaller gains for Llama and Centaur. These findings show that music-mix preference contains systematic listener-, context-, and stimulus-dependent structure while remaining difficult to predict fully. Prior behavioural evidence can improve personalised prediction for selected LLMs, although the evaluated LLMs did not consistently surpass the interpretable mixed-effects approach.*

Keywords— *music mixing, listener preference, personalised audio, Bayesian multilevel modelling, large language models, preference prediction.*

I. INTRODUCTION

Music mixing is both a technical and creative stage of music production in which decisions concerning level, dynamics, spatial balance and spectral relationships shape the perceptual presentation of a recording. The same underlying musical material can therefore support multiple professionally valid mixes reflecting different production choices rather than a single objectively correct solution. Resources such as the Mix Evaluation Dataset demonstrate this variation by providing alternative mixes of the same multitrack material produced by different engineers together with human perceptual evaluations (De Man and Reiss, 2017). Research in intelligent music production has similarly demonstrated the

range of computational approaches through which mixing decisions can be represented and modelled (De Man, Reiss and Stables, 2017).

Although these decisions are made during production, the resulting recording is ultimately encountered and experienced by listeners. Music production therefore also has a relational dimension: alternative production choices may be received differently by different individuals. This does not imply that mixing should be optimised solely for listener preference, but it makes the reception of alternative mixes a meaningful object of study. An important question is consequently whether different listeners systematically prefer different production interpretations of the same musical material.

Listener preference may also be situated rather than stable across circumstances. Music listening serves different functional purposes, including enjoyment, relaxation, distraction, focus and motivation. The Episode Model proposed by Eerola, Kirts and Saarikallio (2025) conceptualises music listening in terms of functional episodes shaped by the listener's goals and circumstances. From this perspective, the characteristics valued by a listener may differ according to what they are using music for. The same individual could therefore prefer one mix in one listening situation and another mix in a different situation. The phenomenon investigated in this dissertation can consequently be framed as variation arising from the relationship between mix, listener and listening context: preference may vary both between listeners and within the same listener across contexts.

This motivates two complementary research problems. The first is empirical: to determine how preferences among alternative professionally produced mixes vary across listeners and contextual listening situations. The second is computational: given such individual and context-dependent variation, to investigate whether an individual listener's preference can be predicted and which sources of information are useful for doing so. The prediction problem involves heterogeneous evidence, including descriptions of listening situations, objective acoustic characteristics, participant-level information, numerical preference ratings and qualitative explanations of previous choices.

Large language models (LLMs) are particularly interesting in this setting because recent research suggests that they can capture regularities in human perceptual and behavioural judgements. Marjeh et al. (2024) demonstrated that LLMs can predict human sensory judgements across several perceptual modalities, while Velissaridis et al. (2025) evaluated foundation models on timbre-related cognitive tasks and their alignment with human perceptual responses. Behaviourally adapted models such as Centaur further suggest that language-model architectures can be specialised for

predicting human decisions from behavioural evidence (Binz et al., 2025). These developments motivate investigating whether LLMs can integrate contextual, acoustic and listener-specific information to predict the preference of a particular listener, rather than only approximate aggregate human judgements.

The present project investigates these questions through a controlled online listening study using professionally produced alternative mixes from the Mix Evaluation Dataset (De Man and Reiss, 2017). Participants evaluate five alternative mixes of musical excerpts under three contextual scenarios representing Enjoyment, Relaxation–Distraction, and Focus–Motivation. For each scenario, participants rate their personal preference for each alternative while imagining themselves in the specified listening situation and provide a comparative qualitative comment. Because the experiment uses only three fixed scenarios, they are treated as controlled contextual conditions rather than as a test of generalisation across arbitrary natural-language listening situations.

Two complementary modelling perspectives are then applied. Bayesian mixed-effects modelling is used to analyse the hierarchical repeated-measures ratings and quantify variation associated with contextual conditions, participants, stimuli and measured acoustic characteristics. LLMs are subsequently evaluated on personalised held-out preference prediction. For a target trial, models receive contextual, acoustic and participant information, with the personalised-history condition additionally providing eligible ratings and qualitative comments from the listener's previous trials. The target-trial response remains withheld. This design allows the study to examine both the structure of human preference variation and the extent to which an individual's preference can be predicted from available evidence.

The study is guided by four research questions that progress from empirical characterisation of listener preference to personalised computational prediction:

RQ1: How do listener, context, stimulus, and acoustic factors relate to preference ratings for alternative music mixes?

RQ2: Can large language models predict an individual listener's preferred music mix on a held-out trial?

RQ3: Does participant history, defined as ratings and qualitative comments from the same listener's previous trials, improve held-out LLM preference prediction?

RQ4: How do personalised LLM predictions compare with a matched mixed-effects predictive model?

The corresponding hypotheses are that preference ratings will exhibit systematic variation associated with listening context alongside meaningful participant- and stimulus-level variability (H1); LLMs will predict an individual listener's preferred mix on held-out trials above the nominal five-class chance level of 20% (H2); incorporating listener-specific ratings and qualitative comments from previous trials will improve prediction relative to prediction without prior-trial preference evidence (H3); and LLM predictive performance will differ from that of the matched mixed-effects statistical predictor (H4). Differences in predictive performance across individual LLMs, listening contexts and musical stimuli are examined as exploratory analyses rather than as a separate research question.

The dissertation makes three connected contributions. First, it provides a controlled listening-study dataset of individual preferences among professionally produced alternative music mixes across functional listening contexts, combining repeated numerical preference ratings with qualitative listener responses. Second, it uses Bayesian multilevel modelling to quantify how these preferences relate to listening context, individual listeners, musical stimuli and measured acoustic characteristics while accounting for the hierarchical repeated-measures structure of the data. Third, it evaluates personalised computational prediction of held-out listener preferences using four LLMs, examining whether prior-trial behavioural evidence improves prediction and comparing personalised LLM performance with a matched mixed-effects predictive model. Together, these contributions connect empirical data collection and statistical characterisation of listener-dependent mix preference with the evaluation of personalised computational prediction.

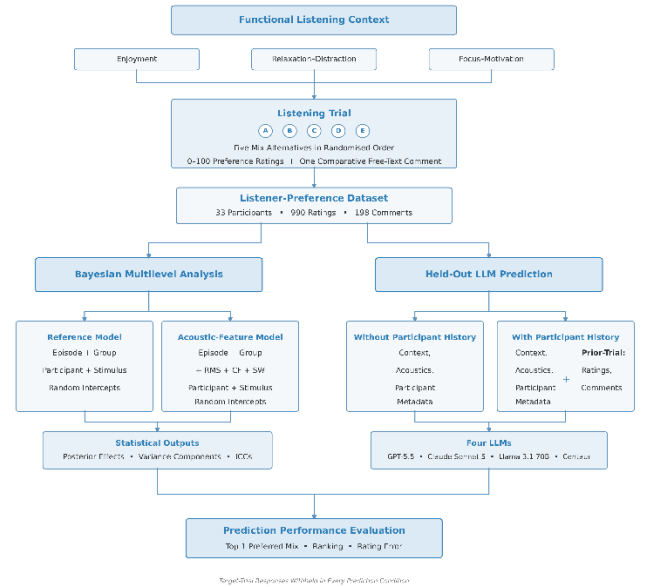


Fig. 1. Overview of the listening study and analysis framework, including preference elicitation, Bayesian multilevel analysis, held-out LLM prediction with and without participant history, and performance evaluation.

II. RELATED WORK

Research in intelligent music production has investigated how mixing decisions can be represented, evaluated and automated computationally. Automatic mixing encompasses a range of approaches for assisting or performing production processes such as level balancing, equalisation, dynamics processing and spatialisation (De Man, Reiss and Stables, 2017). Many such approaches optimise production parameters towards a particular target or solution. The question addressed in the present study is different: rather than generating or optimising a mix, it examines how listeners respond to alternative professionally produced interpretations of the same musical material.

The Mix Evaluation Dataset provides a particularly relevant resource for investigating this question because it contains multiple mixes of the same multitrack recordings produced by different engineers together with human perceptual evaluations (De Man and Reiss, 2017). Alternative mixes therefore reflect ecologically produced differences in production decisions while retaining the same underlying composition and performance. This makes it possible to

examine perceptual responses to production variation without treating one mix as necessarily representing a universally optimal solution.

The existence of multiple production interpretations also motivates consideration of listener-dependent preference. Aggregate evaluations can identify broad tendencies among mixes, but they do not necessarily imply that all listeners prefer the same alternative. The present work therefore uses alternative human-produced mixes to investigate preference at the level of the individual listener and to examine whether those preferences vary under different listening conditions.

Music listening occurs in situations shaped by the listener's activities, goals and desired outcomes. The Episode Model of music listening proposed by Eerola, Kirts and Saarikallio (2025) conceptualises music experiences through functional episodes, providing a framework for considering listening purpose as part of the conditions under which music is experienced. Functions represented within this framework include enjoyment, relaxation, distraction, focus and motivation.

This perspective is relevant to mix preference because the production characteristics valued by a listener need not remain constant across listening situations. A listener seeking enjoyment may not necessarily respond to alternative production balances in the same way when using music to unwind or to maintain engagement during an activity. Preference can therefore be considered in relation to both the individual listener and the functional context in which listening is imagined to occur.

The present experiment operationalises this contextual perspective through three controlled listening scenarios representing Enjoyment, Relaxation–Distraction, and Focus–Motivation. These scenarios are used to investigate whether preferences among alternative mixes vary within listeners across contextual conditions. They are not intended to constitute a broad test of linguistic or semantic generalisation, since only three fixed scenarios are included.

Context has also been considered within intelligent music production. Lefford et al. (2021) argue that mixing decisions are inherently situated and that intelligent mixing systems should account for contextual factors when determining appropriate production choices. Their focus is primarily on the production context surrounding the mix engineer, whereas the present study examines context from the listener side by asking whether preferences among completed mixes vary across functional listening situations.

Recent work has investigated whether foundation models capture regularities associated with human perceptual and behavioural judgements. Marjieh et al. (2024) showed that large language models can predict human sensory judgements across multiple perceptual modalities, suggesting that representations learned from large-scale language data can contain information associated with patterns in human perception even when the models do not directly experience sensory stimuli in the same way as human participants.

Related work in music has examined the extent to which foundation models align with human responses to perceptual tasks. Velissaridis et al. (2025) evaluated several foundation-model families on timbre-related cognitive tasks and investigated their alignment with human perceptual responses. Such findings motivate the use of foundation

models as computational predictors of human judgements rather than assuming that their relevance is limited to conventional natural-language tasks.

This distinction is important for the present study. The objective is not to treat an LLM as a model of the perceptual mechanisms underlying music preference, nor to ask it to evaluate audio directly. Instead, the model is provided with structured representations of the experimental situation and candidate mixes and is evaluated according to its ability to predict observed human preference. The resulting task therefore concerns predictive alignment with listener responses.

Predicting an individual listener introduces a different challenge from reproducing an aggregate human judgement. Individual responses may deviate systematically from population-level tendencies, making information about the particular person potentially valuable for prediction. Language models provide one possible framework for combining heterogeneous evidence about a listener, including structured participant information and natural-language descriptions of previous behavioural responses.

Centaur is particularly relevant to this perspective. Binz et al. (2025) adapted Llama 3.1 70B using a large corpus of human behavioural data with the objective of modelling and predicting human cognition and decision-making. Its relationship to the general-purpose Llama 3.1 foundation provides a useful basis for investigating whether adaptation to behavioural data transfers to a different domain involving individual music-mix preferences.

Previous responses from the same listener may provide a further source of personalisation. Numerical ratings indicate which alternatives the listener previously preferred and by what margin, while qualitative comments may contain information about the characteristics that influenced those preferences. This motivates investigating whether listener-specific behavioural evidence can improve prediction beyond contextual, acoustic and participant-level information alone.

The present study consequently connects two strands of investigation: understanding how preferences among alternative mixes vary across listeners and listening contexts, and predicting those individual preferences with and without access to prior listener-specific behavioural evidence. Foundation models are evaluated as predictors of held-out listener responses, with particular attention to whether prior behavioural evidence supports personalised prediction.

III. EXPERIMENTAL DESIGN

A. Study Overview

A controlled online listening study was used to investigate whether preferences among alternative mixes of the same musical material vary between listeners and across listening contexts. Participants were presented with short contextual vignettes representing different functional listening situations and indicated their personal preference among five alternative mixes while imagining themselves in each situation. By holding the underlying song constant within each comparison, the experiment was designed to focus on differences associated with mixing and production choices rather than composition or performance.

Four songs were included in the experiment and divided between two participant groups. Group 1 evaluated Lead Me

by The DoneFors and I’d Like To Know by Filthybird, while Group 2 evaluated In The Meantime by Fredy V and Pouring Room by The DoneFors. Each participant was assigned to one group and therefore evaluated two songs.

Three contextual listening scenarios were used. Each participant evaluated both assigned songs under each of the three scenarios, producing six experimental trials in total. Within every trial, five alternative mixes of the same song excerpt were presented as anonymous Versions A–E. Participants rated their personal preference for each version on a continuous 0–100 scale while imagining themselves in the specified listening situation. Consequently, every participant provided 30 experimental ratings:

$$3 \text{ contexts} \times 2 \text{ songs} \times 5 \text{ mixes} = 30 \text{ ratings.}$$

In addition to the numerical ratings, participants provided one comparative free-text comment per trial describing the factors that influenced their preferences among the five alternatives. The resulting repeated-measures structure supports analysis of preference variation at the population and participant levels, as well as participant-level held-out preference prediction.

Participants and Ethics. The online listening study was approved by the Queen Mary Research Ethics Committee. Participants were required to be aged 18 years or older; no music production or audio-engineering experience was required. Before beginning the study, participants reviewed the Participant Information Sheet and Consent Form and provided voluntary informed consent through the study interface. No personal identifiers were collected, and responses were stored in anonymised form. All 33 submitted responses were complete, structurally valid and non-duplicate submissions and were therefore retained for analysis.

B. Contextual Listening Scenarios

Three contextual listening scenarios were adapted from the Episode Model of music listening (Eerola, Kirts and Saarikallio, 2025). Rather than treating Enjoyment, Relaxation, Distraction, Focus and Motivation as five independent conditions, the study represents three broader functional contexts: Enjoyment, Relaxation–Distraction, and Focus–Motivation.

The Enjoyment scenario describes listening during free time for entertainment; Relaxation–Distraction describes listening to unwind and temporarily disengage from everyday concerns; and Focus–Motivation describes listening while maintaining engagement during an activity requiring sustained mental or physical effort. These scenarios manipulate the functional purpose of listening while avoiding explicit references to potentially desirable production characteristics such as loudness, stereo width, dynamics or spectral balance, thereby reducing the risk of directly cueing participants towards particular acoustic properties.

Because only three fixed scenarios are used, they are treated as controlled contextual conditions rather than as a sufficiently diverse natural-language sample for evaluating generalisation to arbitrary listening descriptions. The complete participant-facing vignette wording is provided in Appendix A.

C. Stimulus Selection

Candidate stimuli were drawn from the Mix Evaluation Dataset (De Man and Reiss, 2017), which contains multiple

professionally produced alternative mixes of the same multitrack recordings together with historical perceptual evaluations. Four songs were selected for the final experiment, with two assigned to each participant group.

Alternative mixes were characterised using objective acoustic descriptors derived from aligned stereo excerpts. The primary scalar representation comprises root-mean-square energy (RMS), crest factor (CF), stereo width (SW) and stereo imbalance (SI), together with a 48-dimensional mid/side Bark-spectrum representation adapted from Diff-MST (Vanka et al., 2024). RMS, CF and SW are subsequently used as primary interpretable acoustic predictors in the statistical and LLM analyses, while SI is retained primarily for quality control and sensitivity analysis. The Bark representation was used during stimulus selection to capture broader spectral differences but is not included in the primary inferential model.

Stimulus selection combined objective acoustic diversity with historical perceptual information from the source dataset. Candidate alternatives were evaluated under complementary Similar Ratings and Wide Ratings objectives, favouring sets that retained meaningful acoustic separation while representing both relatively similar and more widely separated historical preference scores. Historical ratings were used only to guide stimulus composition and are not treated as ground-truth preferences for participants in the present study.

The resulting six-mix review set for each song was reduced to five alternatives using a final selection criterion combining acoustic separation, historical-rating coverage and avoidance of acoustically near-duplicate alternatives. The selected sets retained a three-plus-two structure, comprising three alternatives with relatively similar historical ratings and two additional alternatives broadening the represented preference range. Full selection functions, scoring weights and intermediate procedures are provided in Appendix B.

Table I summarises the resulting 20 experimental stimuli.

TABLE I. FINAL EXPERIMENTAL STIMULUS SET.

Group	Song	Mix Selection
Group 1	Lead Me – The DoneFors	DU-D, DU-E, PXL-L1, PXL-L4, McG-pro2
	I’d Like To Know – Filthybird	PXL-S1, PXL-S2, PXL-S3, PXL-S5, PXL-S7
Group 2	In The Meantime – Fredy V	DU-H, DU-I, DU-K, QUT-B, QUT-pro
	Pouring Room – The DoneFors	McG-pro1, McG-R, McG-T, McG-V, McG-X

^a. Mix identifiers are researcher-facing labels from the source dataset; participant-facing Version A–E labels are randomised and anonymous.

D. Stimulus Preparation

Because alternative mixes of the same song may contain small timing offsets, candidate mixes were temporally aligned before objective comparison using a two-stage coarse and waveform-based alignment procedure. Alignment quality was verified before a common 28-second excerpt was extracted for each song, ensuring that alternative mixes represented temporally corresponding musical material.

Objective acoustic features were computed from the aligned source excerpts before loudness normalisation and boundary fades. This ensured that descriptors reflected the

original production characteristics rather than transformations introduced for web presentation.

For participant presentation, the final 20 excerpts were rendered as stereo 44.1-kHz WAV files, loudness-normalised to a common target of -20.8 LUFS-I and given short boundary fades. Integrated loudness and true peak were verified after rendering. Full alignment parameters and quality-control procedures are provided in Appendix B.

Controlling excerpt duration and integrated loudness reduces the possibility that preference is driven primarily by duration or global playback level rather than by production differences between mixes.

E. Randomisation and Experimental Control

Randomisation procedures were incorporated to reduce order, position and labelling biases. Participants were assigned to one of the two song groups, each containing two songs. The order of the three contextual scenarios and the presentation order of the songs was randomised. Within each trial, the mapping between the five underlying source mixes and the visible Version A–E labels was also randomised and stored with the participant response.

Participants were therefore not shown the original mix identifiers, source engineer information, historical preference ratings, objective acoustic descriptors or the role that a mix played during stimulus selection. These variables remain researcher-facing metadata used only for analysis.

F. Human Preference Measures

The fundamental human response variable was the continuous 0–100 preference rating assigned to each candidate mix while the participant imagined themselves in the specified listening situation. Each participant contributed five ratings in each of six trials, producing 30 repeated rating observations. These responses preserved information about both the ordering of alternatives and the magnitude of differences in expressed preference.

For participant-level prediction, a preferred-mix outcome was derived within each trial by identifying the version receiving the highest preference rating. This produced a five-class categorical target corresponding to Versions A–E. Tied highest ratings were identified explicitly. For preferred-mix evaluation, a prediction received credit when it matched any candidate belonging to the set of tied highest-rated mixes.

The numerical ratings and derived preferred-mix outcomes supported two related but distinct analyses. Bayesian mixed-effects models used the repeated continuous preference ratings to examine population-level relationships between listening context, stimulus characteristics and participant responses. The LLM evaluation instead focused primarily on predicting the preferred mix for an individual participant on a held-out trial, with prediction of the five numerical ratings treated as a secondary outcome.

Trial-level qualitative comments provided an additional source of listener-specific evidence. In personalised LLM conditions, comments and ratings from a participant's non-target trials were used to characterise that listener's previous preferences, while all response information belonging to the held-out target trial is withheld.

IV. METHODOLOGY

A. Statistical Analysis of Listener Preference

Bayesian multilevel mixed-effects modelling was used as the central empirical analysis to quantify how preference ratings varied with listening context while accounting for repeated observations from the same listeners and evaluations of the same stimuli. The hierarchical structure was represented through random intercepts for participant and stimulus, separating listener-level variability, stimulus-level variability, and residual within-observation variation.

The primary stimulus model was specified as:

$$rating \sim episode + group + (1 | participant_id) + (1 | stimulus_id) \quad (1)$$

where episode represented the listening context and group was included as a fixed design-control factor. Because the two participant groups evaluated different song sets, the group coefficient cannot be interpreted independently as a causal group effect and instead controls for systematic differences associated with the study design.

Variance components from the participant, stimulus, and residual terms were used to quantify the relative contribution of listener- and stimulus-level heterogeneity. Intraclass correlation coefficients (ICCs) were calculated from these variance components to characterise the proportion of rating variability attributable to participants and stimuli.

B. Acoustic Feature Model

A second multilevel model examined whether measurable acoustic characteristics were associated with preference ratings after accounting for listening context and the hierarchical data structure. Three standardised acoustic predictors were included: RMS level (z_RMS), crest factor (z_CF), and stereo width (z_SW). The model was specified as:

$$rating \sim episode + group + z_RMS + z_CF + z_SW + (1 | participant_id) + (1 | stimulus_id) \quad (2)$$

Acoustic features were extracted before loudness normalisation, as described in Section III, preserving differences in the original submitted mixes. Predictors were standardised across the final stimulus set before modelling. Coefficients were interpreted as associations with preference ratings rather than causal effects.

C. Bayesian Estimation and Diagnostics

Both models were estimated within a Bayesian framework using four Markov chain Monte Carlo chains and 1,000 retained posterior draws per chain. Models were implemented using Bambi (Capretto et al., 2022) with the PyMC probabilistic-programming framework (Salvatier, Wiecki and Fonnesbeck, 2016). Posterior estimates were accompanied by credible intervals and convergence diagnostics. Model adequacy was assessed using the Gelman-Rubin R-hat statistic, effective sampling diagnostics, and divergent-transition counts. The final models produced no divergent transitions and a maximum R-hat of 1.01, indicating satisfactory computational convergence.

A simulation-based sample-size analysis had previously been conducted across a range of participant counts and data-generating scenarios to assess expected estimation stability

under the planned hierarchical design. This analysis informed recruitment targets but was not treated as a formal power calculation or as evidence that a particular sample size guaranteed inferential sufficiency; full details are provided in Appendix C.

D. LLM Preference Prediction

Four large language models were evaluated for held-out individual preference prediction: GPT-5.5 (OpenAI, 2026), Claude Sonnet 5 (Anthropic, 2026), Llama 3.1 70B Instruct (Grattafiori et al., 2024), and Centaur (Binz et al., 2025). Quantised implementations were used where required by the available compute infrastructure. Each model was evaluated over the same set of held-out listening trials, with the target trial response excluded from the information supplied to the model.

For each target trial, the model received a structured representation of the listening scenario and the five candidate mixes, including the information available under the corresponding experimental condition. The principal prediction target was the listener's preferred mix. Models additionally produced a ranking of the five candidates and, where supported by the model interface, predicted preference ratings on the study's 0–100 scale. Centaur was evaluated through its native likelihood-based interface and therefore did not produce directly comparable 0–100 ratings; rating-error metrics were consequently not calculated for Centaur.

E. Participant-History Conditions

Each LLM was evaluated under two matched conditions. In the without-participant-history condition, the model predicted the held-out preference without access to responses from that participant's other listening trials. In the with-participant-history condition, the model additionally received ratings and qualitative comments from that same listener's prior trials as participant-specific preference evidence.

Participant history therefore refers specifically to prior-trial ratings and comments from the same participant, rather than to a separate source of biographical or external information. The response to the target trial was excluded in both conditions, preventing direct target leakage. Comparing the two conditions over the same held-out examples tests whether, in this design, prior behavioural evidence improves personalised preference prediction.

F. Prediction Evaluation and Statistical Comparison

Prediction performance was evaluated over the same 198 held-out target trials for each model and condition. The primary classification metric was set-based Top-1 preferred-mix accuracy. Because multiple mixes could share the highest observed human rating within a trial, a prediction received Top-1 credit when the selected mix belonged to the set of tied observed winners. With five candidate mixes per trial, 20% provided the nominal chance reference.

Ranking performance was evaluated using mean Spearman rank correlation between predicted and observed candidate rankings, with ties in the human ratings handled explicitly. For models producing numerical 0–100 preference predictions, mean absolute error (MAE) and root mean squared error (RMSE) were additionally calculated.

The effect of participant history was assessed through paired comparisons between the with- and without-history conditions over the same 198 target trials for each model.

This allowed changes in preferred-mix accuracy, ranking agreement, and rating error to be attributed to the additional prior-trial evidence within the fixed evaluation design.

For comparison with statistical prediction, the primary acoustic mixed-effects model was used as the matched statistical comparator. LLM and mixed-effects predictions were evaluated over the same $N=33$ participant dataset and the same 198 held-out target trials, enabling direct comparison of preferred-mix accuracy and ranking performance and, where compatible numerical predictions were available, rating error. Top-1 uncertainty was quantified using Wilson 95% confidence intervals. Performance against the nominal 20% chance reference was assessed using one-sided exact binomial tests with Benjamini–Hochberg adjustment across the chance-test family. History versus no-history and LLM versus mixed-effects Top-1 comparisons used paired exact McNemar tests on trial-level correctness. Spearman, MAE, RMSE, and paired metric differences used participant-level percentile-bootstrap 95% intervals with 2,000 resamples. All final scoring used the frozen evaluation protocol established before the final comparative analysis.

V. RESULTS

A. Listening Study and Preference Variation

The final analysable dataset comprised 33 participants, with 17 assigned to Group 1 and 16 to Group 2. Each participant completed six listening trials, producing 198 trials and 990 individual mix ratings across the three listening episodes and 20 song-mix stimuli. The Bayesian multilevel models converged satisfactorily, with no divergent transitions and a maximum R-hat of 1.01.

The primary acoustic-feature mixed-effects model showed evidence of systematic variation in preference across listening episodes. Relative to the reference episode EDR-1 (Enjoyment), the estimated coefficient for FM-1 (Focus-Motivation) was -6.64 rating points (95% credible interval $[-10.63, -2.68]$), indicating lower preference ratings in this context after accounting for the remaining model terms. The Group 2 coefficient was 16.77 $[2.03, 32.15]$; however, because participant group was confounded with the song set presented, this coefficient is treated as a design-control association rather than an independent group effect.

In the acoustic-feature model, crest factor was negatively associated with preference ratings ($\beta = -6.27$, 95% credible interval $[-12.44, -0.37]$). The credible intervals for RMS level and stereo width crossed zero, providing less evidence of a consistent association with preference in the present dataset. These coefficients describe associations within the observed stimuli and should not be interpreted as causal acoustic effects.

Variance decomposition further indicated meaningful heterogeneity at both listener and stimulus levels. The participant-level ICC was 0.156 and the stimulus-level ICC was 0.128, while approximately 0.716 of the variance remained at the residual level. Thus, although listening context and measured acoustic characteristics were associated with some systematic variation, preferences also differed meaningfully between listeners and stimuli, with substantial within-observation variability remaining.

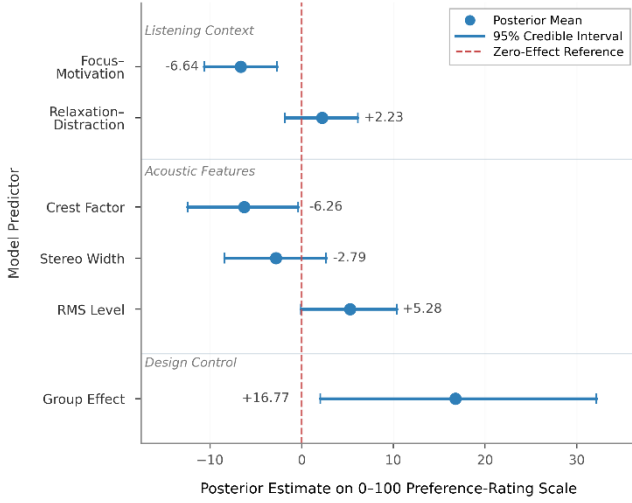


Fig. 2. Posterior coefficient estimates and 95% credible intervals from the primary acoustic-feature Bayesian multilevel model. Enjoyment (EDR-1) and Group 1 serve as the reference levels, and acoustic predictors are standardised. The red dashed line indicates the zero-effect reference.

B. Held-Out Mix Preference Prediction

Held-out preferred-mix prediction was evaluated over the same 198 target trials, with 20% representing the nominal chance level for selection among five candidate mixes. Performance was modest overall but exceeded this reference for the strongest model configurations. Among the LLMs, Claude Sonnet 5 with participant history achieved the highest Top-1 accuracy at 35.9%, followed by GPT-5.5 with participant history at 34.3%. The matched mixed-effects predictor also achieved 34.3% Top-1 accuracy.

Performance differed across both models and information conditions, indicating that predictive success was not uniform across LLM architectures. Ranking agreement with the observed preference ordering was also evaluated using tie-aware Spearman correlation, while MAE and RMSE quantified numerical rating error for models producing comparable 0–100 predictions. The mixed-effects predictor achieved the strongest headline ranking agreement (mean Spearman $\rho = 0.347$) and an RMSE of 27.59, demonstrating that the statistical model remained a competitive predictor despite the additional representational flexibility of the LLM approaches.

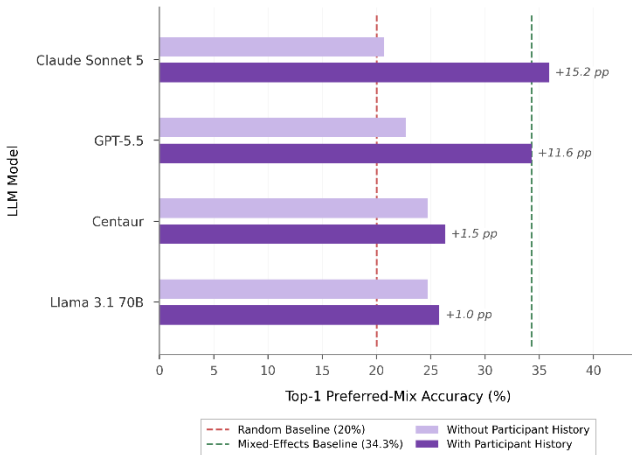


Fig. 3. Top-1 preferred-mix accuracy for the four LLMs with and without participant-history evidence. The red dashed line indicates the nominal random baseline (20%), and the green dashed line indicates the matched

mixed-effects baseline (34.3%). Values beside the with-history bars show the change in Top-1 accuracy relative to the corresponding without-history condition, in percentage points (pp).

C. Effect of Participant History

Providing prior-trial participant evidence produced strongly model-dependent changes in held-out prediction. For GPT-5.5, Top-1 accuracy increased by 11.6 percentage points relative to the corresponding condition without participant history, while Claude Sonnet 5 improved by 15.2 percentage points. In contrast, the gains for Llama 3.1 70B Instruct and Centaur were substantially smaller, at 1.0 and 1.5 percentage points, respectively.

These results indicate that prior ratings and qualitative comments from the same listener can improve personalised preference prediction, but that the benefit is not universal across models. The substantial gains observed for GPT-5.5 and Claude Sonnet 5 therefore support the usefulness of prior listener-specific evidence for some LLMs, while the limited changes for Llama and Centaur caution against generalising the effect to LLM-based prediction as a whole.

D. Comparison with the Mixed-Effects Predictive Model

Direct comparison over the matched 198 held-out trials showed that personalised LLM prediction did not clearly outperform the mixed-effects statistical predictor. GPT-5.5 with participant history matched the mixed-effects model at 34.3% Top-1 accuracy (difference = 0.0 percentage points; $p = 1.000$), while Claude Sonnet 5 with participant history reached 35.9%, 1.5 percentage points higher than the mixed-effects predictor; this numerical difference was not statistically detectable ($p = .822$). The mixed-effects model additionally achieved the strongest headline ranking agreement (mean Spearman $\rho = 0.347$).

The comparison therefore provides no evidence that the evaluated personalised LLMs consistently surpassed the matched statistical approach. Instead, the strongest LLM configurations reached broadly comparable preferred-mix prediction accuracy, while the mixed-effects model remained competitive in ranking performance. These results highlight the value of participant-specific information for selected LLMs without displacing the statistical model as a strong and interpretable predictor of listener preference.

VI. CONCLUSION

This dissertation investigated how preferences for alternative music mixes vary across listeners and functional listening contexts, and whether these individual preferences can be predicted computationally. A controlled listening study collected 990 preference ratings across 198 trials from 33 participants, covering four songs, five alternative mixes per song, and three listening contexts. Bayesian multilevel modelling provided the central empirical analysis, while four LLMs were subsequently evaluated for held-out preference prediction with and without prior listener-specific evidence.

The statistical analysis showed that music-mix preference was not constant across the experimental conditions. Listening context was associated with preference ratings, with the Focus-Motivation episode receiving lower ratings than the reference Enjoyment episode. Crest factor was also negatively associated with preference within the evaluated stimulus set, whereas the estimated associations for RMS level and stereo width were less conclusive. Participant- and stimulus-level variance further demonstrated meaningful

heterogeneity between listeners and mixes. At the same time, substantial residual variability remained, illustrating the difficulty of explaining subjective mix preference from the measured factors alone.

Held-out prediction results showed that individual preference was predictable to a modest degree. The strongest LLM configuration, Claude Sonnet 5 with participant history, achieved 35.9% Top-1 preferred-mix accuracy against a nominal five-choice chance reference of 20%, while GPT-5.5 with participant history achieved 34.3%. Importantly, the value of listener-specific evidence was model-dependent. Providing prior-trial ratings and comments improved Top-1 accuracy by 15.2 percentage points for Claude and 11.6 percentage points for GPT-5.5, but produced only small improvements for Llama 3.1 70B Instruct and Centaur. Prior behavioural evidence can therefore support personalised prediction, but the effect should not be assumed to generalise uniformly across LLMs.

Comparison with the matched mixed-effects predictor provides an important qualification to these findings. The statistical model achieved 34.3% Top-1 accuracy and the strongest headline ranking agreement (mean Spearman $\rho = 0.347$). GPT-5.5 with participant history matched its Top-1 accuracy, while Claude was numerically 1.5 percentage points higher; however, this difference was not statistically detectable. The results therefore provide no evidence that the evaluated LLMs consistently surpassed the statistical approach. Instead, the two modelling paradigms offered complementary perspectives: multilevel modelling provided an interpretable account of systematic preference variation, while selected LLMs demonstrated an ability to exploit qualitative and behavioural listener history for personalised prediction.

The work makes three main contributions. First, it provides a controlled listener-preference dataset combining repeated numerical ratings and qualitative responses across professionally produced alternative mixes and functional listening contexts. Second, it demonstrates a Bayesian multilevel framework for quantifying context-, listener-, stimulus-, and acoustic-related variation in music-mix preference. Third, it provides a frozen held-out evaluation of four LLMs under matched personalised and non-personalised conditions, together with a direct comparison against a statistical predictor over the same target trials.

Several limitations constrain the generality of the findings. The final sample comprised 33 participants and four songs divided between two participant groups, limiting the diversity of listeners and musical material represented. Group was consequently confounded with song set and cannot be interpreted as an independent causal effect. The listening episodes were imagined functional contexts rather than observations of listening behaviour in naturally occurring environments. Preference ratings are inherently subjective, and considerable residual variation remained after accounting for the measured factors. In addition, the statistical and LLM approaches do not have identical representational capabilities, and Centaur's native likelihood-based interface prevented direct comparison using 0–100 rating-error metrics.

Future work should evaluate larger and more diverse listener populations and musical material, including whether listener characteristics such as musical or audio-production

experience systematically moderate mix preference and context-dependent preference variation. Richer hierarchical models could examine listener-specific contextual or acoustic relationships where larger datasets support their estimation, while prediction studies could evaluate generalisation to previously unseen listeners and stimuli. Audio-capable multimodal models may additionally provide a more direct representation of the musical signal than the text and acoustic-feature representations considered here. More broadly, the findings suggest that personalised music-mix prediction is feasible but remains challenging, and that future systems are likely to benefit from combining interpretable statistical modelling with richer listener-specific evidence rather than treating either approach as a complete account of subjective preference.

ACKNOWLEDGMENTS

Special thanks to all participants who took part in the listening study, whose time and responses made the empirical analysis presented in this dissertation possible.

REFERENCES

- Anthropic (2026) ‘Introducing Claude Sonnet 5’, 30 June. Available at: <https://www.anthropic.com/research/claude-sonnet-5> (Accessed: 16 August 2026).
- Binz, M., Akata, E., Bethge, M., Brändle, F., Callaway, F., Coda-Forno, J., Dayan, P., Demircan, C., Eckstein, M.K., Gureckis, T.M., Ji-An, L., Mattar, M.G., Mollick, E., Palminteri, S., Pleskac, T.J., Rahwan, I., Schulz, E., Speekenbrink, M., Summerfield, C. and Wulff, D.U. (2025) ‘A foundation model to predict and capture human cognition’, *Nature*, 644, pp. 1002–1009.
- Capretto, T., Piho, C., Kumar, R., Westfall, J., Yarkoni, T. and Martin, O.A. (2022) ‘Bambi: A simple interface for fitting Bayesian linear models in Python’, *Journal of Statistical Software*, 103(15), pp. 1–29. doi: 10.18637/jss.v103.i15.
- De Man, B. and Reiss, J.D. (2017) ‘The Mix Evaluation Dataset’, in *Proceedings of the 20th International Conference on Digital Audio Effects (DAFx-17)*, Edinburgh, UK, 5–9 September 2017.
- De Man, B., Reiss, J.D. and Stables, R. (2017) ‘Ten Years of Automatic Mixing’, in *Proceedings of the 3rd Workshop on Intelligent Music Production (WIMP 2017)*, Salford, UK, 15 September 2017.
- Eerola, T., Kirks, C. and Saarikallio, S. (2025) ‘Episode model: The functional approach to emotional experiences of music’, *Psychology of Music*, 53(4). doi: 10.1177/03057356241279763.
- Grattafiori, A. et al. (2024) ‘The Llama 3 Herd of Models’, arXiv preprint, arXiv:2407.21783.
- Lefford, M.N., Bromham, G., Fazekas, G. and Moffat, D. (2021) ‘Context-Aware Intelligent Mixing Systems’, *Journal of the Audio Engineering Society*, 69(3), pp. 128–141. doi: 10.17743/jaes.2020.0043.
- Marjeh, R., Sucholutsky, I., van Rijn, P., Jacoby, N. and Griffiths, T.L. (2024) ‘Large language models predict human sensory judgments across six modalities’, *Scientific Reports*, 14, 21445. doi: 10.1038/s41598-024-72071-1.
- OpenAI (2026) ‘Introducing GPT-5.5’, 23 April. Available at: <https://openai.com/index/introducing-gpt-5-5/> (Accessed: 16 August 2026).
- Salvatier, J., Wiecki, T.V. and Fonnesbeck, C. (2016) ‘Probabilistic programming in Python using PyMC3’, *PeerJ Computer Science*, 2, e55. doi: 10.7717/peerj-cs.55.
- Vanka, S., Steinmetz, C., Rolland, J.-B., Reiss, J. and Fazekas, G. (2024) ‘Diff-MST: Differentiable Mixing Style Transfer’, in *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR 2024)*, San Francisco, USA.
- Velissaridis, G., Athwal, R., Musharaf, M., Fazekas, G. and Saitis, C. (2025) ‘Evaluating foundation models on timbre-related cognitive tasks’, in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR 2025)*, Daejeon, South Korea.

APPENDIX A - LISTENING STUDY MATERIALS

A. Contextual Listening Vignettes

Enjoyment — EDR-1: Enjoying Free Time

“You have some free time and no immediate responsibilities that require your attention. You are not feeling tired or worried, but you would like to enjoy your time rather than let it pass uneventfully. You decide to listen to music for entertainment, making your free time more enjoyable.”

Relaxation–Distraction — EDR-2: Unwinding After a Demanding Period

“You have some time for yourself after a demanding period. Although a few everyday concerns are still on your mind, there is nothing that requires your attention right now. You would like to feel calmer and take your mind off these concerns for a while, so you decide to listen to music to help you unwind.”

Focus–Motivation — FM-1: Maintaining Engagement During an Activity

“You are engaged in an activity that requires sustained mental or physical effort. As you continue, it becomes more difficult to stay engaged. You want to maintain your focus and motivation so that you can continue making steady progress toward completing the activity. You decide to listen to music while carrying on with the activity.”

APPENDIX B - STIMULUS SELECTION AND AUDIO PREPARATION

A. Excerpt Alignment Procedure

Alternative mixes of each song were aligned before objective acoustic comparison. For alignment, audio was temporarily downmixed to mono and resampled to 11,025 Hz. A coarse alignment representation combined frame-level root-mean-square energy with positive spectral flux, and candidate offsets within ± 2 s were compared using normalised dot-product correlation.

A subsequent waveform-based refinement searched within ± 0.20 s of the coarse estimate using 5 ms increments. Alignment confidence was assessed against a predefined threshold of 0.7. Following alignment, a common 28-second excerpt was identified for each song so that all candidate mixes contained temporally corresponding musical material.

Objective feature extraction was performed on the aligned excerpts before loudness normalisation and boundary fades.

B. Similar and Wide Rating Triplet Selection

Objective acoustic diversity was combined with historical perceptual information from the Mix Evaluation Dataset. Existing preference scores were matched to candidate mixes and aggregated separately within each song.

Candidate triplets were evaluated under two complementary objectives, termed Similar Ratings and Wide Ratings. For each triplet, an acoustic score combined normalised minimum pairwise distance and mean pairwise distance with weights of 0.7 and 0.3, respectively. Rating information favoured low historical-rating spread for the Similar Ratings condition and high spread for the Wide Ratings condition.

The combined triplet-selection objective assigned a weight of 0.8 to acoustic diversity and 0.2 to historical-rating structure, with additional coverage and quality-control terms. The Wide Ratings triplet was selected to minimise overlap with the highest-scoring Similar Ratings triplet. The two

triplets were then combined to produce a six-mix review union for each song.

C. Final Five-Mix Selection

All five-mix subsets of each six-mix review union were evaluated using a dedicated final-set criterion. The scoring function considered minimum acoustic separation, median acoustic separation, historical-rating range, preservation of a similar-rating subset and avoidance of acoustically near-duplicate alternatives.

The final-set score assigned weights of 0.38 to normalised minimum acoustic distance, 0.25 to normalised median acoustic distance, 0.22 to normalised historical-rating range and 0.15 to the similar-rating term, with a penalty of 0.5 for each identified near-duplicate pair.

Within each candidate five-mix set, the closest-rating triplet was identified as the similar-rating subset, while the remaining alternatives served as range-broadening or bridge mixes. These roles were recomputed after final selection rather than inherited directly from the original Similar Ratings and Wide Ratings triplets.

APPENDIX C – STATISTICAL ANALYSIS DETAILS

A. Sample-Size Design Analysis

A simulation-based design analysis was conducted before participant recruitment to assess the approximate number of analysable participants required for stable Bayesian estimation. Candidate sample sizes of $N = 20, 30, 40, 50, 60$ and 80 were examined under three synthetic effect and noise scenarios.

Although 20 replications per sample-size/scenario combination were initially planned, this would have required 360 repeated Bayesian fits and was reduced for computational practicality to 10 replications per condition. The executed simulation therefore comprised 180 shortened-chain fits, using two chains, 400 tuning draws, 400 retained posterior draws and a target acceptance rate of 0.95, with no failed fits. A separate confirmation fit was subsequently conducted for the candidate $N = 50$ design using four chains, 1,000 tuning draws and 1,000 retained posterior draws.

The simulations were developed under an earlier factorial version of the experiment. The final listening study subsequently adopted a five-mix, three-context structure while retaining 30 ratings per participant. The simulation results are therefore interpreted as design-stage recruitment guidance rather than as a formal power calculation for the final experimental specification.

The final documented recommendation was $N = 50$ analysable participants. Recruitment beyond this number was considered desirable to accommodate incomplete responses, exclusions and other unusable observations. Larger sample sizes, including $N = 60$, were also investigated and generally provided additional precision, but $N = 60$ was not retained as the final minimum recommendation.

Final inference was based on posterior estimates and uncertainty from the collected human data rather than on the synthetic simulations alone. The final $N=33$ models converged satisfactorily, with no divergent transitions and a maximum R-hat of 1.01.

APPENDIX D – LLM EVALUATION PROTOCOL

A. Prompt Structure and Input Conditions

The primary evaluation used zero-shot prompting. No separate few-shot demonstrations were included, and explicit chain-of-thought reasoning was not requested. Each prompt contained a structured description of the prediction task, the target listening context, the candidate Version A–E information permitted by the experimental condition, and the relevant participant information.

Two principal conditions were evaluated. The non-history condition excluded behavioural outcomes from the participant's previous trials. The personalised-history condition additionally provided eligible non-target trials from the same participant, including previous preference ratings, comparative qualitative comments and corresponding trial contexts. All target-trial response information remained withheld in both conditions.

B. Structured Output Schema

For models supporting the common structured prediction interface, the required response contained three prediction components:

predicted_preferred_mix: predicted highest-preference candidate among Versions A–E;

predicted_ratings: predicted numerical preference ratings for the five candidates on the 0–100 study scale;

predicted_ranking: predicted ordering of Versions A–E.

A free-text rationale was not required as part of the primary response schema. Centaur was evaluated through its native likelihood-based interface rather than the common 0–100 rating interface; consequently, directly comparable numerical rating predictions and associated MAE/RMSE metrics were not available for Centaur.

C. Parsing and Invalid-Output Handling

Model responses were parsed against the predefined structured-output specification. If an initial response could not be parsed into the required format, one formatting-only repair was permitted, without introducing new substantive reasoning or additional task information. Invalid or missing outputs that remained unresolved were retained in the denominator of the primary evaluation rather than being silently excluded.

D. Held-Out Evaluation and Leakage Controls

For each target trial, all response-derived information belonging to that trial was excluded from the model input. This included the participant's five observed preference ratings, qualitative comment, derived preferred-mix outcome and any other variables derived from those responses.

In the personalised-history condition, only eligible non-target trials from the same participant were permitted to provide behavioural evidence. Target outcomes were not permitted to influence outcome-dependent preprocessing or normalisation. The prediction task therefore used information available independently of the response being predicted.

Participant-facing Version A–E identifiers were treated as arbitrary candidate labels. Their mapping to underlying stimuli was retained in the evaluation records so that predictions could be compared with the corresponding observed responses without exposing the target outcome to the model.

E. Model and Inference Configuration

Final production inference evaluated GPT-5.5, Claude Sonnet 5, Llama 3.1 70B Instruct, and Centaur. GPT-5.5 and Claude Sonnet 5 were evaluated through their corresponding hosted model interfaces, while open-weight models were executed using the available research compute infrastructure, with quantisation applied where required by computational constraints. Llama 3.1 70B Instruct used a 4-bit NF4 implementation.

The final production identifiers were gpt-5.5 (deployment revision gpt-5.5-2026-04-23), claude-sonnet-5, and meta-llama/Llama-3.1-70B-Instruct. Centaur used the marcelbinz/Llama-3.1-Centaur-70B-adapter adapter with a 4-bit Llama 3.1 70B base. Exact checkpoint revisions and execution metadata were retained with the frozen experiment artifacts.

F. Experiment Logging and Reproducibility

Experiment records preserve the model identity, input condition, prompt content, anonymised participant and trial identifiers, candidate-label mapping, raw model response, parsed prediction and evaluation status. Together with the frozen prompt specification and evaluation scripts, these records support reconstruction and auditing of the final prediction experiments.

APPENDIX E – DETAILED RESULTS

A. Mixed-Effects Model Results

Table II reports the posterior fixed-effect estimates from the stimulus reference model and the primary acoustic-feature model. Estimates are expressed on the 0–100 preference-rating scale.

TABLE II. BAYESIAN MULTILEVEL MODEL RESULTS

Model	Quantity	Estimate	95% CrI
Stimulus Model	Intercept	50.302	[40.279, 60.146]
	Relaxation–Distraction	2.202	[−1.860, 5.874]
	Focus–Motivation	−6.675	[−10.641, −2.893]
	Group Effect	5.995	[−7.258, 18.819]
Primary Acoustic-Feature Model	Intercept	44.800	[34.622, 55.207]
	Relaxation–Distraction	2.231	[−1.798, 6.130]
	Focus–Motivation	−6.641	[−10.627, −2.677]
	Group Effect	16.766	[2.029, 32.145]
	RMS (z)	5.284	[−0.073, 10.389]
	Crest factor (z)	−6.265	[−12.443, −0.369]
	Stereo width (z)	−2.795	[−8.416, 2.679]

a. Enjoyment and Group 1 are the reference levels. The group effect represents Group 2 relative to Group 1. Acoustic features are standardised (z). Bold indicates non-intercept effects whose 95% credible intervals exclude zero.

The primary acoustic-feature model produced a participant ICC of 0.156 (95% CrI [0.084, 0.239]), a stimulus ICC of 0.128 [0.044, 0.218], and a residual variance share of 0.716 [0.613, 0.813]. Corresponding values for the stimulus model were 0.145 [0.075, 0.219], 0.173 [0.082, 0.279], and 0.682 [0.572, 0.780], respectively.

B. Prediction Results

Table III reports the complete held-out prediction results over the matched 198 target trials.

TABLE III. HELD-OUT PREFERENCE PREDICTION RESULTS

Model	Condition	Top-1	95% CI	Chance-test p (BH)	Spearman	MAE	RMSE
GPT-5.5	Without participant history	22.70%	[0.174, 0.291]	0.215	−0.021	29.77	35.74
	With participant history	34.30%	[0.281, 0.412]	5.13×10^{-6}	0.325	21.75	30.22
Claude Sonnet 5	Without participant history	20.70%	[0.156, 0.269]	0.430	−0.092	29.90	35.26
	With participant history	35.90%	[0.295, 0.427]	1.43×10^{-6}	0.248	24.47	31.73
Llama 3.1 70B	Without participant history	24.70%	[0.193, 0.312]	0.077	0.006	30.43	36.56
	With participant history	25.80%	[0.202, 0.323]	0.052	0.039	29.56	35.50
Centaur	Without participant history	24.70%	[0.193, 0.312]	0.077	0.002	N/A	N/A
	With participant history	26.30%	[0.206, 0.328]	0.044	0.013	N/A	N/A
Mixed-effects Baseline	Baseline	34.30%	[0.281, 0.412]	5.13×10^{-6}	0.347	23.00	27.59

b. History denotes prior-trial ratings and comments from the same participant. Bold indicates the best LLM performance for each metric (higher is better for Top-1 and Spearman; lower is better for MAE and RMSE). Chance-test p-values compare Top-1 accuracy with the nominal five-choice random baseline of 20% and are Benjamini–Hochberg adjusted. MAE and RMSE are not applicable to Centaur because it does not produce calibrated 0–100 preference ratings.

APPENDIX F – GENERATIVE AI TOOLS: STUDENT ACCOUNTABILITY STATEMENT

Student Name: Oscar Alejandro Gallegos Villarreal

Project Title: Individual Listener Preferences for Music Mixes Across Listening Contexts: Statistical Analysis and Personalised LLM Prediction

Supervisor Name: Georgios Velissaridis

1. Have you used Generative AI tools for your project?

YES

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
ChatGPT	5.5	OpenAI	https://chatgpt.com/
Claude	Sonnet 5	Anthropic AI	https://claude.com/
Llama	3.1 70B Instruct	Meta	https://developer.meta.com/ai/docs/overview/
Centaur	Llama 3.1 Centaur 70B	Binz et al. / Helmholtz Munich	https://huggingface.co/marcelbinz/Llama-3.1-Centaur-70B-adapter

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Project Section	Description	Justification
Project Planning and Experimental Workflow	Was used to support planning and organisation of the project phases.	Used as a planning and organisational aid. Final methodological decisions and implementation were reviewed and determined by the researcher.
Listening-Study Interface Development	Was used during development of the web-based study interface, including generation, implementation of code, debugging, and randomization of data.	Accelerated software development and debugging. Generated or suggested code was reviewed, tested and validated before deployment.
Data Processing and Cleaning	Assisted with developing and reviewing scripts used to parse, validate, restructure and clean responses collected through the listening-study interface.	Reduced implementation and debugging time while maintaining a reproducible scripted data-processing pipeline. Final data and exclusion/validation outputs were checked by the researcher.
Code Auditing	Was used to review code, identify potential errors or bugs, suggest corrections and assist with debugging across the statistical-analysis and LLM-experiment pipelines.	Provided additional review. Suggested changes were inspected and tested rather than accepted automatically.
LLM Preference-Prediction Experiments	GPT-5.5, Claude Sonnet 5, Llama 3.1 70B Instruct and Centaur were directly evaluated as predictive models in the dissertation experiments using the predefined held-out evaluation protocol.	Use of these AI models was an explicit component of the research methodology and was necessary to answer the dissertation research questions concerning personalised preference prediction.
Repository Auditing	Was used to inspect the project repository, inconsistencies, missing outputs, and discrepancies between the implemented experiments and methodology.	Provided an additional quality-assurance step. Final claims and numerical results were verified against the repository's experiment artifacts rather than relying only on AI summaries.

Dissertation Proofreading and Editing	Was used to identify grammatical errors, and inconsistencies.	Used to improve readability and language quality. Scientific claims, interpretations and final wording were reviewed by the researcher and checked against the underlying results and cited literature.
Figures and Presentation	Was used to help design and refine dissertation figures.	Used to improve presentation of results; the underlying data, values and scientific interpretation were derived from the project's verified outputs.

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: Oscar Alejandro Gallegos Villarreal Date: 19 / August / 2026